

Exercise: SNAT and Prethreaded Server

Problem

A client in a private LAN connects to an echo server through an SNAT router:

```
Client: 192.168.1.10:10086
SNAT public mapping: 192.168.1.10:10086 <=> 202.120.40.82:233
Server: 202.120.40.85:15213
```

The server is a prethreaded concurrent server. A master thread calls `Accept()` and inserts each connected descriptor into a bounded buffer. Worker threads remove descriptors from the buffer, call `echo(connfd)`, and close the connection.

Answer the following questions.

Part 1. Sockets Interface

Briefly explain `connect`, `bind`, `listen`, and `accept`. What is the difference between `listenfd` and `connfd`? What socket pair does the server see after SNAT?

Part 2. SNAT

Fill in the source and destination socket addresses:

1. Client to router, before SNAT.
2. Router to server, after SNAT.
3. Server to router, before reverse translation.
4. Router to client, after reverse translation.

Why does the server not see `192.168.1.10:10086` directly?

Part 3. Bounded Buffer Synchronization

Design the bounded descriptor buffer used between the master thread and workers. Give the needed semaphores, their initial values, and the `P` / `V` order in:

```
void sbuf_insert(sbuf_t *sp, int connfd);
int sbuf_remove(sbuf_t *sp);
```

Also explain why the `P` order matters.

Part 4. Prethreading

Why can a prethreaded server perform better than:

1. An iterative server.
2. A thread-per-connection server.

Mention both concurrency and resource overhead.

Part 5. Performance Trade-offs

The server uses N worker threads and a buffer of size B . Discuss what can go wrong if N or B is too small or too large.

Part 6. Fairness and Starvation

Does this semaphore-based FIFO buffer guarantee fairness among clients or worker threads? Could starvation still happen? Briefly explain.